// Day 26 of 30

COMPARISON

RabbitMQ vs Kafka — Choose Wisely

→ Swipe through all 6 slides

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Core Difference

Message broker vs event stream

```
RabbitMQ = Smart Broker
  → Routes messages to consumers
  → Message deleted after consumed
  → Traditional messaging

Kafka = Dumb Broker, Smart Consumer
  → Append-only log
  → Messages persist (configurable)
  → Consumers track their position
```

💡 RabbitMQ delivers. Kafka stores.

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When RabbitMQ Wins

Task queues, routing, simple async

```
Best for:
  → Send email after signup
  → Process payment async
  → Route to specific consumer
  → Request-reply pattern
  → Simple pub-sub

Examples:
  → Notification service
  → Background job processing
  → Order → kitchen routing
```

💡 RabbitMQ = simple, reliable, easy to set up

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When Kafka Wins

High throughput, event sourcing, streaming

```
Best for:
  → Millions of events/second
  → Event sourcing (replay events)
  → Stream processing
  → Multiple consumers same data
  → Analytics pipeline

Examples:
  → User activity tracking
  → Real-time analytics
  → Log aggregation
```

💡 Kafka = massive scale, event history, streaming

Performance

Numbers comparison

```
Throughput:
  RabbitMQ: ~50,000 msg/sec
  Kafka:    ~1,000,000 msg/sec

Latency:
  RabbitMQ: ~1ms (lower)
  Kafka:    ~5ms (higher)

Message retention:
  RabbitMQ: deleted after consume
  Kafka:    kept for days/weeks
```

💡 Need speed? Kafka. Need low latency? RabbitMQ.

Complexity

Setup and maintenance

```
RabbitMQ:
  → Easy to set up (5 minutes)
  → Web management UI built-in
  → Good documentation
  → Single binary

Kafka:
  → Needs Zookeeper (or KRaft)
  → More ops knowledge needed
  → Partition management
  → Steeper learning curve
```

💡 Small team? Start with RabbitMQ. Scale? Add Kafka later.

The Decision Framework

Simple flowchart

```
Need replay of old events?
  → Kafka

Need complex routing?
  → RabbitMQ

Millions of events/second?
  → Kafka

Simple async tasks?
  → RabbitMQ

Multiple consumers same event?
  → Kafka

First time using message queues?
  → RabbitMQ
```

💡 Most backend apps start with RabbitMQ. That's fine.

```
done = true
```

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